US Bank Credit Stress Test

Macro satellite model on public FDIC data

A macro stress test on ten large US banks using public FDIC call reports, FRED macro history, and the 2025 Fed severely adverse scenario. This report walks through the model, validation, and stress projections.

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1. How banks stress credit (and what this project does)

In a large bank, credit stress testing usually runs through probability of default (PD), loss given default (LGD), and exposure at default (EAD). Under IFRS 9, expected credit loss (ECL) combines those three at the loan or facility level, with ratings migration and Stage 1, 2, 3 allocation driving provisions. In CCAR and DFAST, banks build segment models for CRE, cards, mortgages, and C&I, often with collateral and prepayment detail, all shocked under a supervisory macro scenario.

That bottom-up stack is the production system. It needs loan tapes, internal ratings, write-off history, and full model validation every cycle.

This project is not that. It is a top-down satellite model: NPL ratio regressed on lagged NPL and a small set of macro variables (GDP, unemployment, mortgage rates). Losses are backed out with a fixed LGD. Capital is equity over assets with a flat PPNR assumption. No risk-weighted assets, no AFS marks, no management actions.

I built it on public data on purpose. The workflow still matches what I used at the State Bank of Pakistan running quarterly macro stress tests across 52 banks: scenario, satellite model, validation, balance-sheet projection. The simplification is in the data and loss engine, not in the logic.

2. The workflow

1. Build a panel: FDIC financials merged with FRED macros (2005 Q1 to 2024 Q4).
2. Estimate a satellite NPL model with bank fixed effects.
3. Validate: in-sample backtest, 2023-2024 out-of-sample forecast, diagnostics.
4. Stress test: 2025 Fed severely adverse scenario to NPL paths, losses, capital.

Each step is a separate Python module. You can read the scenario loader, regression, validation flags, and capital formulas independently.

3. Data

Ten banks: JPMorgan, BofA, Wells Fargo, Citi, U.S. Bank, PNC, Truist, Fifth Third, KeyBank, Regions. Data from the FDIC BankFind API (free, no key). NPL ratio = NCLNLS / LNLSNET x 100. A PD model would track rating migration. Here I model the portfolio NPL outcome directly.

CERT	Bank
628	JPMorgan Chase NA
3510	Bank of America NA
3511	Wells Fargo NA
7213	Citibank NA
6548	U.S. Bank NA
6384	PNC Bank NA
9846	Truist Bank
5649	Fifth Third Bank NA
6672	KeyBank NA
18409	Regions Bank

780 bank-quarters after lags (10 banks x 80 quarters minus lag rows). Stress scenario: 2025 Fed severely adverse, nine quarters from 2025 Q1.

4. The satellite model

NPL(i,t) = a_i + rho*NPL(i,t-1) + b1*GDP(t-2) + b2*UR(t) + b3*Mtg(t-1) + error. Pooled OLS with bank dummies and HC1 robust standard errors. Bank FE only, no time FE.

I arrived at four macro terms by dropping lags and variables step by step. GDP at lag 2, contemporaneous unemployment, mortgage rate at lag 1. HPI, CPI, and T-bill dropped from the final spec.

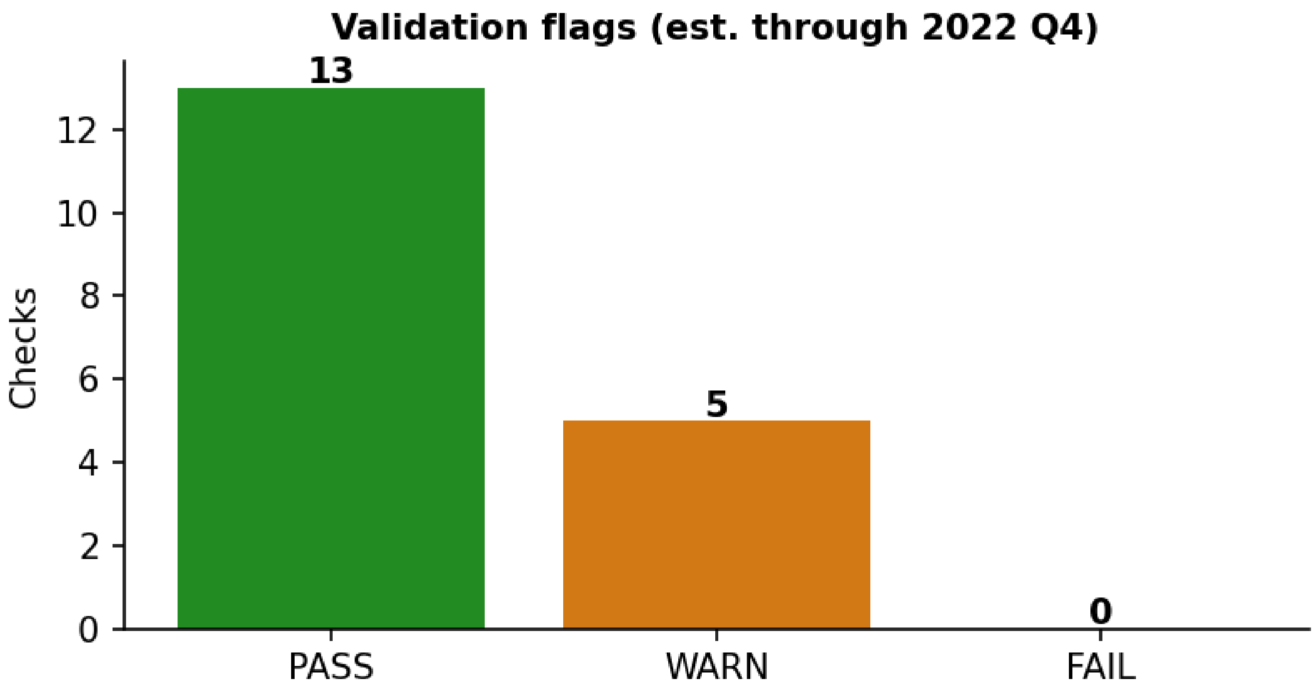
Variable	Coef.	Std err	Sign	Reading
npl_lag1	0.942	0.011	+	High persistence
real_gdp_growth_lag2	-0.007	0.002	-	Weaker growth raises NPL
unemployment	0.052	0.010	+	Higher UR raises NPL
mortgage_rate_lag1	0.077	0.008	+	Higher rates raise NPL

R-squared = 0.972. All coefficients significant at 1%. Signs match credit-cycle theory.

5. Model validation

The validate_model() function produces PASS, WARN, and FAIL flags. Summary: 12 PASS, 4 WARN, 0 FAIL (estimation through 2022 Q4).

Figure 1. Automated validation check counts.



Check	Status	Detail
sign npl lag1	PASS	coef=0.9396, expected +
significance npl lag1	PASS	p-value=0.0000
sign real gdp growth lag2	PASS	coef=-0.0065, expected -
significance real gdp growth lag2	PASS	p-value=0.0002
sign unemployment	PASS	coef=0.0502, expected +
significance unemployment	PASS	p-value=0.0000
sign mortgage rate lag1	PASS	coef=0.0962, expected +
significance mortgage rate lag1	PASS	p-value=0.0000
r squared	PASS	R^2=0.9715
durbin watson	WARN	DW=1.1182
ljung box	WARN	residual autocorrelation detected
residual normality	WARN	Jarque-Bera p=0.0000
heteroskedasticity	PASS	Breusch-Pagan p=0.0000, HC1 robust SEs used in estimation
multicollinearity	WARN	high VIF (>10): unemployment
in sample rmse	PASS	RMSE=0.3204 pp, MAE=0.1915 pp
gfc period rmse	WARN	2008-2009 RMSE=0.7491 pp
out of sample rmse	PASS	OOS RMSE=0.1726 pp (0.54x in-sample)
out of sample bias	PASS	OOS bias=0.1445 pp

WARN flags: serial correlation (DW = 1.12), non-normal residuals (GFC fat tails), unemployment VIF = 10.7, GFC RMSE = 0.75 pp. These are documented, not hidden.

6. Does the model track realized NPL?

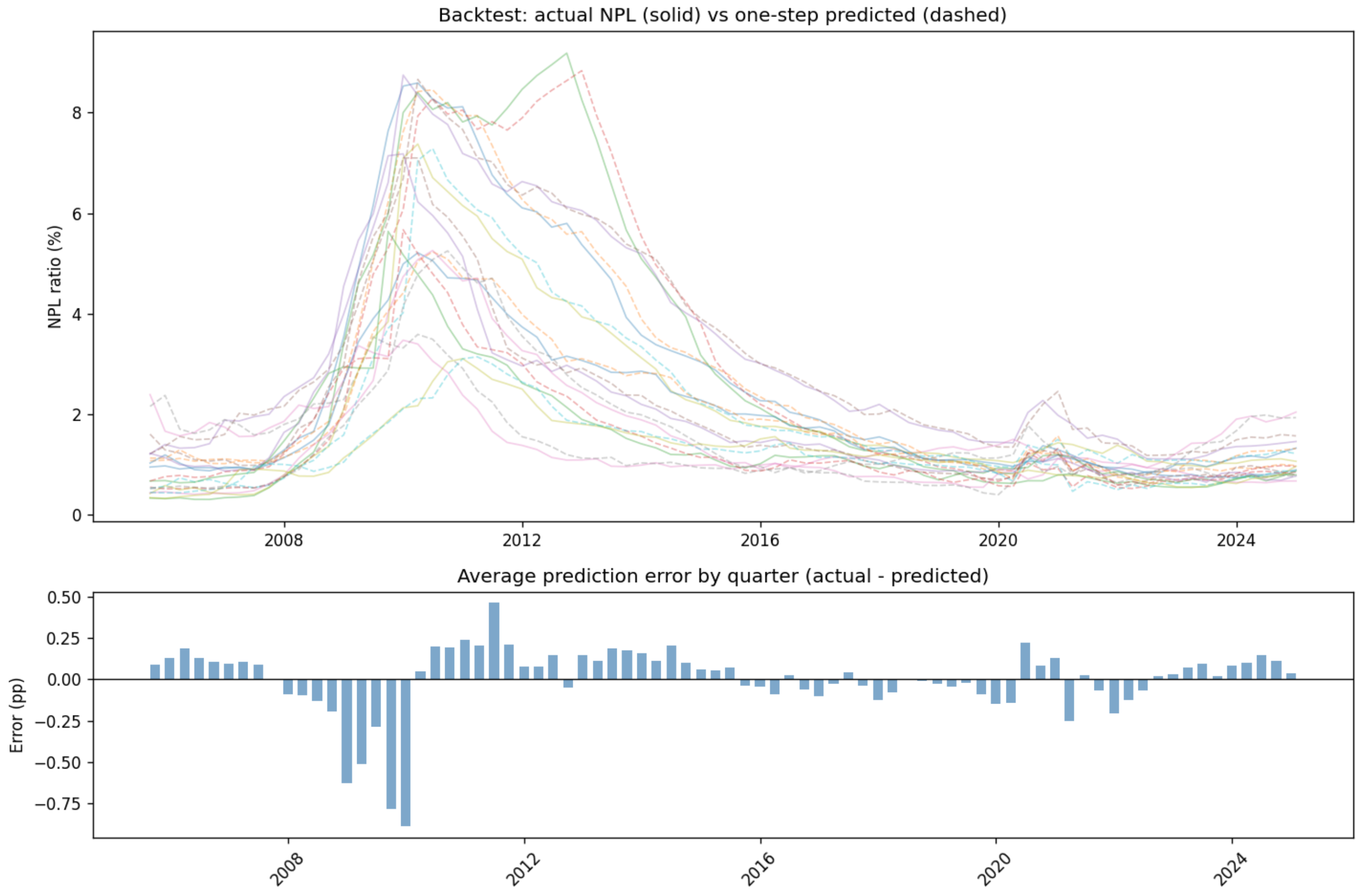
In-sample backtest

One-step-ahead prediction at each quarter using only prior information. Overall RMSE = 0.31 pp, MAE = 0.18 pp (percentage points of NPL ratio).

Bank	RMSE	MAE
Regions	0.14	0.11
U.S. Bank	0.19	0.13
Fifth Third	0.23	0.18
Truist	0.24	0.16
Wells Fargo	0.33	0.19
JPMorgan Chase	0.34	0.20
Citibank	0.35	0.21
KeyBank	0.35	0.18
Bank of America	0.40	0.26
PNC	0.40	0.18

Period	RMSE (pp)
2005-2007	0.20
2008-2009	0.75
2010-2014	0.29
2015-2019	0.11
2020-2022	0.18

Figure 2. Top: actual NPL (solid) vs one-step predicted (dashed) by bank. Bottom: average prediction error by quarter. Errors spike in the GFC.

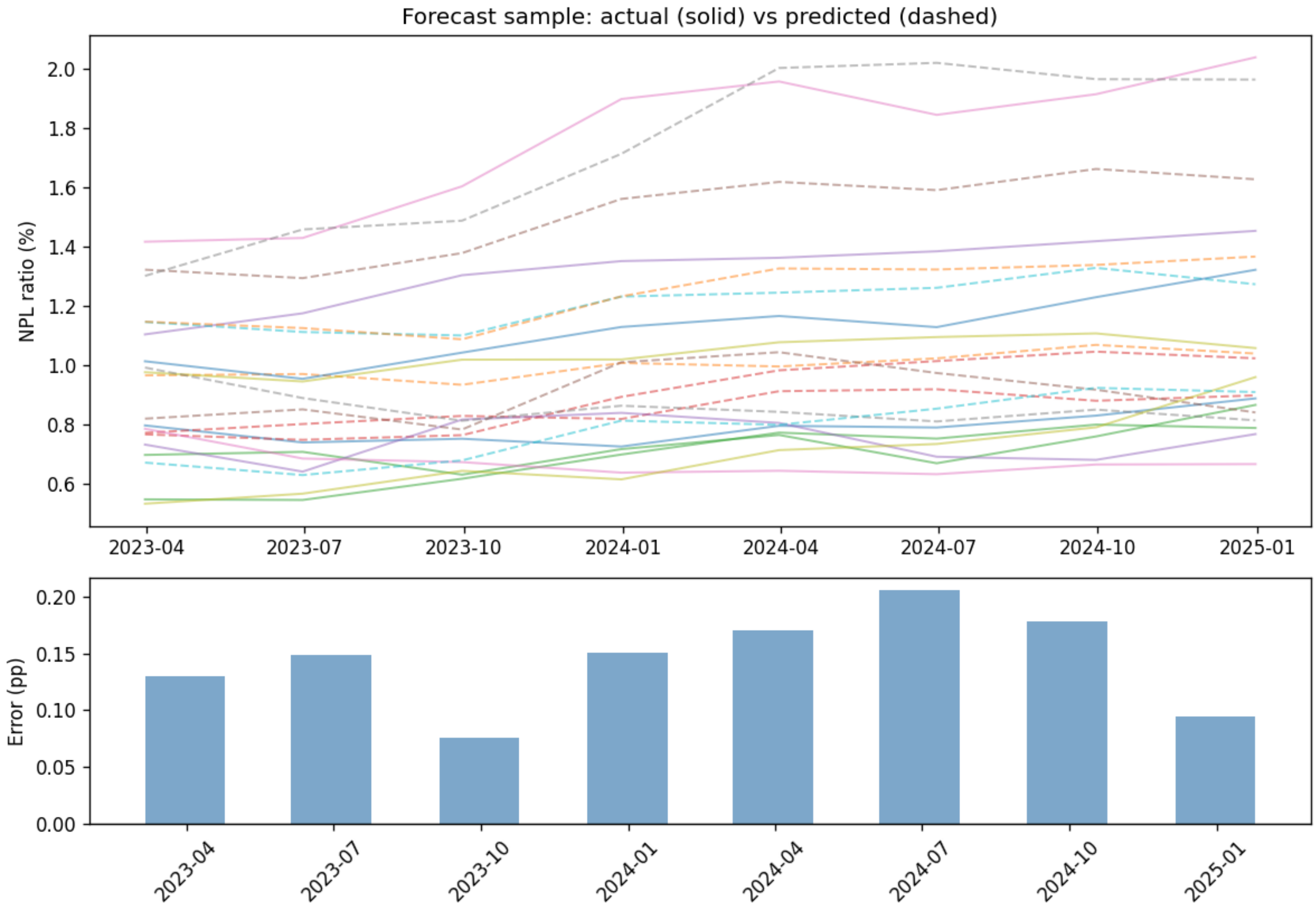


Out-of-sample forecast (2023 Q1 to 2024 Q4)

Model estimated through 2022 Q4 only. Holdout: 2023-2024. RMSE = 0.17 pp, MAE = 0.16 pp, bias = +0.14 pp (model under-predicts realized NPL).

Quarter	Actual %	Pred %	Error pp
2023 Q1	0.86	0.99	+0.13
2023 Q2	0.84	0.99	+0.15
2023 Q3	0.91	0.98	+0.08
2023 Q4	0.96	1.11	+0.15
2024 Q1	1.01	1.18	+0.17
2024 Q2	0.97	1.18	+0.21
2024 Q3	1.02	1.20	+0.18
2024 Q4	1.08	1.17	+0.09

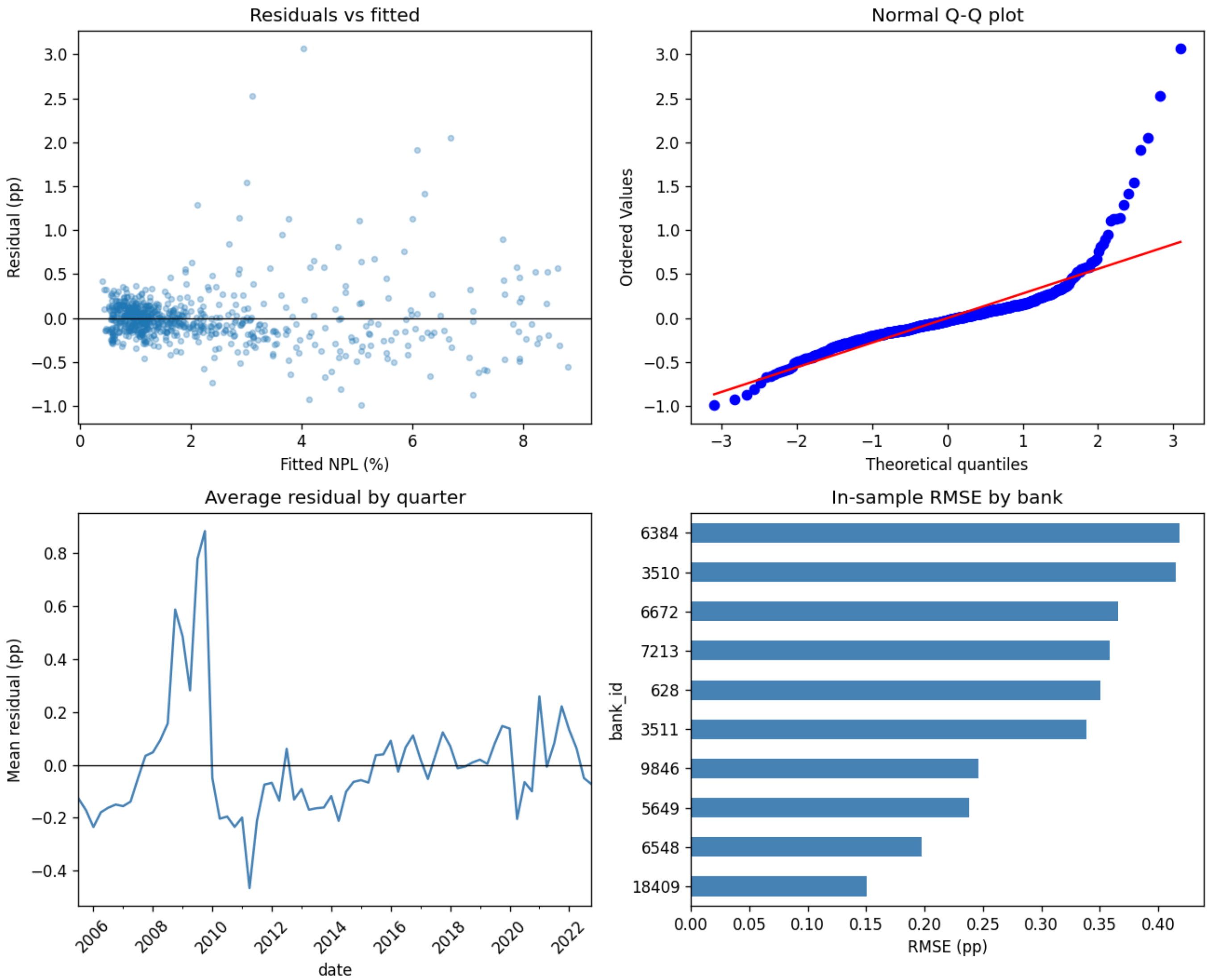
Figure 3. Out-of-sample window. Lines track more closely than in the GFC. Bars show the model lagging the 2024 NPL uptick.



Regression diagnostics

Durbin-Watson = 1.12 (autocorrelation). Jarque-Bera rejects normality (GFC tails). Breusch-Pagan rejects homoskedasticity (HC1 SEs used). Unemployment VIF = 10.7.

Figure 4. Top left: residuals vs fitted. Top right: Q-Q plot (fat upper tail). Bottom left: mean residual by quarter (GFC spike). Bottom right: RMSE by bank.

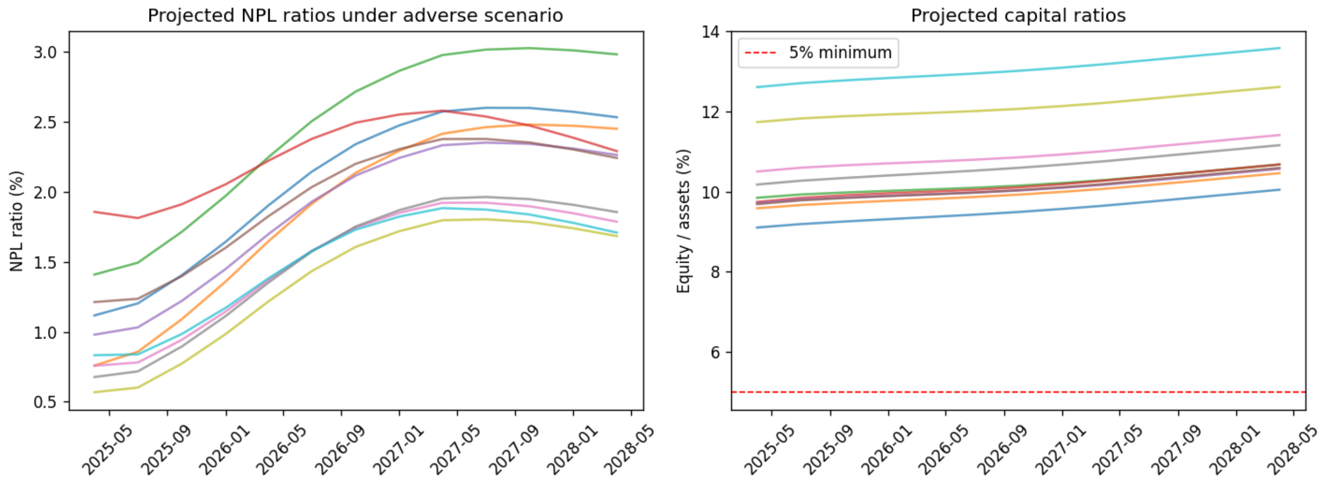


7. Stress test results

2025 Fed severely adverse scenario from 2024 Q4 positions. Losses = delta NPL x net loans x LGD (45%). PPNR = 0.10% quarterly ROA under stress. Capital = equity / assets. No RWA, no dividends, no management actions.

Bank	Trough cap	End cap	Credit loss	Breach 5%?
JPMorgan Chase	9.10%	10.04%	$10.3B	No
Bank of America	9.58%	10.46%	$8.4B	No
U.S. Bank	9.68%	10.59%	$2.0B	No
PNC	9.71%	10.56%	$1.9B	No
Fifth Third	9.74%	10.67%	$0.4B	No
Wells Fargo	9.84%	10.67%	$6.3B	No
Citibank	10.17%	11.15%	$3.7B	No
KeyBank	10.50%	11.41%	$0.6B	No
Truist	11.73%	12.61%	$1.7B	No
Regions	12.60%	13.57%	$0.8B	No

Figure 5. Left: NPL roughly doubles under the adverse macro path. Right: equity/assets ratios. Capital rises slightly because PPNR exceeds losses in this simplified module. That is a limitation, not a solvency finding.



8. Limitations and disclaimer

This is not a PD/ECL engine. Capital is not CET1. PPNR is a flat ROA. The linear model misses GFC nonlinearities. Ten banks is a demo sample.

Educational project on public data only. Not supervisory software. Outputs are not assessments of any real bank.

9. Author

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GitHub: github.com/muhammadumarshinwari/us-bank-credit-stress-test